Projection and Anti-Concentration Certificates for the Low-Temperature SK Fluctuation Scale

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M399

October 9, 2026

Abstract

The supplied manuscript claims the one-sixth fluctuation exponent for the zero-field Sherrington–Kirkpatrick log partition function at each fixed inverse temperature greater than one. This report studies two quantitative interfaces in that argument. An optimized allocation of field-clock lengths removes the logarithmic loss in its one-branch projection estimate, without changing its conditional-law requirements or using a growing cavity recursion. An exact inverse identity for one-sided exponential smoothing yields an interval anti-concentration bound for the original free energy, uniformly in the interval center and down to fixed widths. The latter uses no auxiliary common Gaussian. Explicit positive-Laplace, variance, and approximation-error budgets state what actual finite certificates would have to establish. The global hierarchy, profile rigidity, and propagation arguments remain separate upstream inputs; neither a finite moment table nor a successful rational fixture certifies them. Contribution assessment: C3/5, the loss-free projection refinement and sharp exponential-unsmoothing interface give notable source-specific quantitative improvements without a new global fluctuation exponent or limiting-law theorem.

1 Model and scope

Fix $1 < \beta < \infty$, put $T = \beta^2$, and let $(g_{ij})_{i<j}$ be independent standard real Gaussians. We use

$$H_n(\sigma) = \frac{\beta}{\sqrt{n}} \sum_{i<j} g_{ij}\sigma_i\sigma_j, \qquad F_n = \log\left(2^{-n} \sum_{\sigma\in\{-1,1\}^n} e^{H_n(\sigma)}\right).$$

There is no external field. Every unordered pair is counted once. Disorder expectation is denoted by $\mathbb{E}$. The source [1] claims, for each fixed $\eta > 0$ and sufficiently large $n = n(\beta, \eta)$,

$$n^{1/6-\eta} \le \sqrt{\operatorname{Var} F_n} \le n^{1/6+\eta}, \qquad \mathbb{P}\{n^{1/6-\eta} \le |F_n - \mathbb{E}F_n| \le n^{1/6+\eta}\} \longrightarrow 1.$$

Its conclusion is a power exponent, not a variance prefactor or a limiting distribution. We do not import the companion limiting-law claim as a proved input. Nothing below is uniform as $\beta \downarrow 1$ or $\beta \to \infty$.

For replicas define $R(\sigma, \tau) = n^{-1}\sigma \cdot \tau$. The exact covariance is

$$\mathbb{E}H_n(\sigma)H_n(\tau) = \frac{nT}{2}R(\sigma, \tau)^2 - \frac{T}{2}.$$

An independent $D \sim N(0, T/2)$ therefore gives the completed free energy $F_n^c = F_n + D$. With $K_Y(t) = \log \mathbb{E} e^{t(Y - \mathbb{E}Y)}$,

$$K_{F_n^c}(t) = K_{F_n}(t) + Tt^2/4, \qquad \operatorname{Var} F_n^c = \operatorname{Var} F_n + T/2. \tag{1}$$

Changing the spin prior to counting measure adds $n \log 2$; dividing by n, or multiplying by the fixed physical factor $-1/\beta$, has the corresponding deterministic effect on scales.

The local results proved below concern the source's actual positive planted laws and an exact distributional smoothing operation. Classical Gaussian integration by parts, convexity, entropy minimization, and weak convolution identities retain their classical attribution. The new increment is their quantitative use at these particular interfaces, not a renamed general matrix formalism.

2 The hierarchy and its physical covariances

For nondecreasing interaction and field clocks b, h on mass $u \in [0, 1]$, the source uses features

$$\phi_b(\sigma) = \frac{\sigma\sigma^{\mathsf{T}}}{\sqrt{2n}}, \qquad \phi_h(\sigma) = \sigma, \qquad y = \sigma/\sqrt{n}.$$

Their covariance kernels are $nR^2/2, nR$. At a finite partition the continuation recursion is

$$F_{j-1}(z) = \mathcal{T}_{m_j} F_j(z) = \frac{1}{m_j} \log \mathbb{E} e^{m_j F_j(z + Z_j)}. \tag{2}$$

At mass zero this means ordinary expectation. The forward transition density is exactly $\exp\{m_j(F_j - F_{j-1})\}$ relative to the corresponding Gaussian increment, hence has mass one. A deterministic split tree is built by sharing specified prefixes and then drawing independent continuations. Before and after a jump are different specified cuts. A joint filling of simultaneous clock jumps must be fixed.

At a prefix $\mathcal{F}_u$, put

$$X_u = \mathbb{E}[y \mid \mathcal{F}_u], \quad C_u = \mathbb{E}[yy^{\mathsf{T}} \mid \mathcal{F}_u] - X_u X_u^{\mathsf{T}}, \quad S_u = \mathbb{E}|X_u|^2.$$

For two independent continuations from that prefix,

$$B_u = \mathbb{E} R_{12}^2 = \mathbb{E} \left\| C_u + X_u X_u^{\mathsf{T}} \right\|_{\mathrm{HS}}^2, \qquad D_u^2 = B_u - S_u^2. \tag{3}$$

Consequently, with no commutation assumption,

$$D_u^2 = \operatorname{Var}(|X_u|^2) + 2\mathbb{E} X_u^{\mathsf{T}} C_u X_u + \mathbb{E} \operatorname{tr} C_u^2.$$

The Hessian H_u of a continuation in a source coupled to the normalized spin y satisfies $uC_u \preceq H_u \preceq C_u$. Indeed, differentiating the normalized recursion gives

$$\nabla \mathcal{T}_m F = \mathbb{E}_m \nabla F, \qquad \nabla^2 \mathcal{T}_m F = \mathbb{E}_m \nabla^2 F + m \operatorname{Cov}_m(\nabla F).$$

Iteration represents the Hessian by conditional martingale covariance increments weighted by their masses, including weight one for the terminal spin draw. All remaining masses lie in $[u, 1]$. Freezing the normalizing denominator would omit the covariance term. The original normalized-kernel proof [8] makes precisely this distinction.

The source's pinned Brascamp–Lieb moment estimate is for a full, unconditioned single planted path and its terminal spin. It is not a moment bound uniform over every realized prefix, and not an inequality after arbitrary spin-dependent reweighting. The projection argument below uses the actual normalized continuation law, not that invalid stronger interpretation.

3 Removing a logarithmic projection loss

Consider a branch b and a vector V determined by exterior branches and information at or before their fork. Conditional on that information, the future of b must have its usual planted law. Let a subsequent field-clock interval have length $D_h > 0$ and all its masses be at least $v > 0$. Its beginning and end are deterministic clock cuts, with their jump sides or fillings specified. Fix an even integer $p \geq 2$, and assume $\|V\|_p < \infty$. Write

$$Y = V \cdot y_b, \qquad Y_t = V \cdot X_{b,t}, \qquad a_t = \|Y - Y_t\|_p .$$

All norms include the exterior law. The source's Lemma comp : projection bounds a_{end} by a constant times $(\log n)^2 \|V\|_p / (v\sqrt{nD_h})$. We retain the same integration-by-parts identity but change the deterministic allocation of clock lengths.

Lemma 3.1 (One-interval inequality)**.** *For any subinterval of field length $d > 0$,*

$$a_{\mathrm{end}}^p \leq \frac{B_p \|V\|_p}{v\sqrt{nd}}\, a_{\mathrm{start}}^{p-1}. \tag{4}$$

One may take $B_2 = 1$; for even $p \geq 4$, the safe choice

$$B_p = 2^{\,2p-2}\gamma_p, \qquad \gamma_p = (\mathbb{E}|Z|^p)^{1/p}, \quad Z \sim N(0,1),$$

suffices. The constant is independent of the number and lengths of clock subdivisions.

Proof. Condition first on the exterior information. The normalized planted field driver has innovation $\mathrm{d}G - t\sqrt{n}X_t\ \mathrm{d}h(t)$. Gaussian integration by parts, including every continuation normalization, gives, for $P = (Y - Y_{\mathrm{start}})^{p-1}$,

$$\mathbb{E}\left[P \int_I V \cdot (\,\mathrm{d}G - t\sqrt{n}X_t\ \mathrm{d}h(t))\right] = \sqrt{n}\int_I \mathbb{E}\left[P\left\{t(Y - Y_t) + \int_t^1 (Y - Y_z)\ \mathrm{d}z\right\}\right]\ \mathrm{d}h(t).$$

This is the source's one-branch identity; it holds first on finite Gaussian fillings and passes to its specified bounded-clock limit. It does not condition on the terminal spin.

For z after the start, conditional centering removes $(Y_z - Y_{\mathrm{start}})^{p-1}(Y - Y_z)$. The elementary monotonicity inequality for the odd power $p - 1$ therefore gives

$$\mathbb{E}[P(Y - Y_z)] \geq 2^{2-p}\mathbb{E}|Y - Y_z|^p \geq 0.$$

For z between start and end, conditional contraction implies $a_{\mathrm{end}} \leq 2a_z$. Retain just the first term of the integration-by-parts right side, use $t \geq v$, and discard the nonnegative later-mass integral. Its value is at least $\sqrt{n}\, d\, v\, 2^{2-2p} a_{\mathrm{end}}^p$. The innovation on the left is conditionally Gaussian with variance $d|V|^2$. Its L^p norm is $\gamma_p\sqrt{d}\ \|V\|_p$. Holder's inequality proves (4). For $p = 2$, orthogonal projection gives $a_z \geq a_{\mathrm{end}}$, and the monotonicity constant is one, so $B_2 = 1$. □

Lemma 3.2 (Optimal finite clock allocation)**.** *Suppose a nonnegative endpoint quantity obeys (4) on every finite subdivision of a fixed interval of length D_h. Put*

$$r = 1 - \frac{1}{p}, \quad A = \frac{B_p \|V\|_p}{v\sqrt{n}}, \quad q_i = (1 - r)r^{N-i}, \quad S_N = 1 - r^N.$$

For N subintervals, the product upper bound is minimized by

$$d_i = D_h \frac{q_i}{S_N}, \qquad 1 \leq i \leq N. \tag{5}$$

If $a_0 < \infty$, then

$$a_{\mathrm{end}} \le a_0^{r^N} \left(\frac{A}{\sqrt{D_h}} \right)^{S_N} \exp\left\{ -\frac{1}{2} \sum_{i=1}^{N} q_i \log \frac{q_i}{S_N} \right\}. \tag{6}$$

Proof. Taking the p-th root in each one-step inequality and iterating gives $a_N \le a_0^{r^N} A^{S_N} \prod_i d_i^{-q_i/2}$. For positive lengths with sum D_h, minimizing the logarithm amounts to minimizing $-\sum q_i \log(d_i/D_h)$. The weighted arithmetic–geometric mean inequality, or the nonnegativity of relative entropy, gives its unique minimum at $d_i/D_h = q_i/S_N$. Substitution gives the formula. Zero quantities follow directly or by a positive regularization. □

Theorem 3.3 (Projection without a logarithmic loss)**.** *Under the stated one-branch hypotheses,*

$$\|V \cdot (y_b - X_{b,\mathrm{end}})\|_p \le \frac{B_p g_p \|V\|_p}{v\sqrt{nD_h}}, \qquad g_p = \sqrt{p} \left(1 - \frac{1}{p}\right)^{-(p-1)/2} \le \sqrt{ep}. \tag{7}$$

In particular the $p = 2$ constant can be taken to be 2. Differences of two later conditional projections satisfy the same bound up to a factor two.

Proof. At fixed n and fixed branch, $a_0 \le 2\|V\|_p < \infty$. Apply (6) for every integer N. The endpoint is unchanged. As $N \to \infty$,

$$\sum_{i=1}^{N} q_i \log(q_i/S_N) \longrightarrow \log(1-r) + \frac{r}{1-r} \log r.$$

Thus the last factor tends to $(1-r)^{-1/2} r^{-r/[2(1-r)]} = g_p$. Also $a_0^{r^N} \to 1$ when $a_0 > 0$, and $S_N \to 1$. The case $a_0 = 0$ is immediate. The bound $(1 - 1/p)^{-(p-1)} \le e$ follows from $\log(1+x) \le x$. Conditional contraction and the triangle inequality give the assertion about later projections. □

The limiting allocation is an analytic estimate at fixed n, not an n-dependent cavity expansion. It introduces no new replica labels and no higher moment order. Every finite subdivision has positive length; the one-interval identity has constants independent of that length. Thus neither a uniform lower bound on all subinterval lengths nor a growing-depth hierarchy assertion is being inferred from a finite computation. A coarse physical discretization that cannot realize these fillings must use the finite bound (6) and account for its discretization error instead.

For $p = 2, N = 3$, the clock fractions are $1/7, 2/7, 4/7$, with the largest interval last. For $n = 400, v = 1/2, D_h = 1$ and $\|V\|_2 = 1$, (7) gives $a_{\mathrm{end}} \le 1/5$. This is an implication from a specified planted-law geometry, not an observed finite SK error.

3.1 A specific downstream saving

Retain the source's regular-band layer geometry in comparisons.tex, including its raw bound $\|\Delta_i\|_{4l} \le C_l Q \sqrt{w_i}$. For every pair except the first pair, the chosen preceding field interval has length at least $c\rho v e(v)/\log n$, masses at least cw_k, and an exterior anchor of norm at most $C_l Q\sqrt{w_k}$. The improved projection bound, followed by the same conditional Jensen step, gives

$$\mu_j \left\| \Delta_i^a \cdot \Delta_j^b \right\|_{2l}^2 \le \frac{C_l (\log n) Q^2}{\rho n v e(v)}.$$

Here $\mu_j \lesssim w_k$, so the scale factors cancel. There are $O((\log n)^2)$ pairs and one further $O(\log n)$ Cauchy–Schwarz layer factor. After integration over mass $O(v)$, their total is bounded by

$$\frac{C_l(\log n)^4 Q^2 v^4}{\mathsf{B}_v \rho}, \qquad \mathsf{B}_v = ne(v)v^4. \tag{8}$$

This replaces the source's displayed $(\log n)^8$ bound for those pairs; the remaining $\log n$ comes from the actual field-clock lower bound and has not disappeared. The first-pair estimate, cap/floor geometry, and polynomial saving requirements remain unchanged. No improved final power exponent or temperature range is claimed from this intermediate refinement.

4 The original free energy and exponential augmentation

The common Gaussian in (1) is essential to the source's quadratic covariance convention, but need not be added when studying the distribution of F_n itself. This observation is not a new augmentation principle: the transitive Gaussian formulation of Aronow–Lopatto [2, Proposition 5.1] already includes the off-diagonal model. We give the short argument to keep the precise normalization visible.

Proposition 4.1 (Augmentation without Gaussian completion)**.** *Let E_1 be independent exponential of rate one, and set $Z_n = F_n + E_1$. Its density is log-concave. For $0 < t < 1$,*

$$\begin{aligned} K_{Z_n}(t) &= K_{F_n^c}(t) - Tt^2/4 - t - \log(1-t), \\ \sigma_n^2 &:= \operatorname{Var} Z_n = \operatorname{Var} F_n + 1. \end{aligned} \tag{9}$$

Proof. Write $F_n(g) = \log(2^{-n}\sum_\sigma e^{a_\sigma \cdot g})$ using only off-diagonal disorder coordinates. The density is

$$q(z) = e^{-z}\mathbb{E}\big[e^{F_n(g)}\mathbf{1}_{\{F_n(g)\le z\}}\big].$$

For every spin vector τ, the orthogonal transformation $g_{ij} \mapsto \tau_i\tau_j g_{ij}$ preserves F_n and Gaussian measure and permutes the spin labels transitively. Expanding e^{F_n}, all 2^n integrals are equal. Thus q is the marginal of

$$(2\pi)^{-d/2}\exp(-|g|^2/2 + a_{\sigma_*}\cdot g - z)\mathbf{1}_{\{z\ge F_n(g)\}}.$$

The epigraph is convex since F_n is convex. The integrand is log-concave, and it is integrable because integration in z leaves $e^{a_{\sigma_*}\cdot g - F_n(g)} \le 2^n$ times Gaussian density. The classical Prékopa marginal theorem [5] applies. Independence, $\mathbb{E}e^{t(E_1-1)} = e^{-t}/(1-t)$, and (1) give the identities. □

The rate one is important for this proof: it cancels the partition function after pinning. Arbitrary rates will be permitted in the next convolution identity, but log-concavity for those rates is not inferred. Neither arbitrary spin-dependent reweighting nor external field is covered by this transitivity argument.

5 Sharp one-sided unsmoothing

For a real random variable Y, define its closed-interval concentration function $Q_Y(h) = \sup_{a\in\mathbb{R}} \mathbb{P}\{a \le Y \le a+h\}$, $h \ge 0$. No density of Y is assumed.

Theorem 5.1 (Exact inverse and sharp interval bound)**.** *Let μ be a probability measure on $\mathbb{R}$, and let q_λ be the density of $\mu * \mathrm{Exp}(\lambda)$, $\lambda > 0$, with the right-continuous representative*

$$q_\lambda(z) = \lambda \int_{(-\infty, z]} e^{-\lambda(z-x)}\, \mu(\,\mathrm{d}x).$$

Then q_λ is locally of bounded variation and, as measures,

$$\mu = q_\lambda(z)\,\mathrm{d}z + \lambda^{-1} D q_\lambda.$$

In particular, for every finite $a \le b$,

$$\mu([a,b]) = \int_a^b q_\lambda(z)\,\mathrm{d}z + \frac{q_\lambda(b) - q_\lambda(a-)}{\lambda}. \tag{10}$$

Consequently

$$Q_Y(h) \le \min\{1, (h + \lambda^{-1})\, \|q_\lambda\|_\infty\}. \tag{11}$$

The coefficient $h + \lambda^{-1}$ cannot be decreased in this class, even if the smoothed density is required to be log-concave.

Proof. The function $e^{\lambda z} q_\lambda(z)$ is a locally finite nondecreasing Stieltjes integral, with differential $\lambda e^{\lambda z} \mu(\,\mathrm{d}z)$. The distributional product rule gives $D q_\lambda = \lambda \mu - \lambda q_\lambda\,\mathrm{d}z$. Integration over $[a,b]$ gives (10), including its left limit at a. At $a = b$, the identity says $\mu(\{a\}) = [q_\lambda(a) - q_\lambda(a-)]/\lambda$. Since $q_\lambda \ge 0$, discarding the negative endpoint term proves the bound. The supremum of this representative equals its essential supremum: after a jump it cannot decrease faster than exponential decay, so a large point value is seen on a positive-length interval.

For sharpness, translate the interval to $[0, h]$ and take

$$M = (h + \lambda^{-1})^{-1}, \qquad \mu = \frac{M}{\lambda}\delta_0 + M \mathbf{1}_{(0,h)}(x)\,\mathrm{d}x.$$

This is a probability measure. Direct convolution gives zero for $z < 0$, $q_\lambda(z) = M$ for $0 \le z \le h$, and $q_\lambda(z) = M e^{-\lambda(z-h)}$ for $z > h$. Its log slope changes from zero to $-\lambda$, so it is log-concave, and equality holds in (11). The case $h = 0$ is a single atom. These extremizers are not asserted to be SK free-energy distributions. □

For example, at $\lambda = 1, h = 2$, the measure $\frac{1}{3}\delta_0 + \frac{1}{3}\mathbf{1}_{(0,2)}\,\mathrm{d}x$ has interval mass one and smoothed maximum $1/3$. Formula (10) reads $1 = 2/3 + 1/3$. Omitting the atom at the left endpoint would instead give $2/3$. There is no extra factor two and no smoothing-tail probability error to balance against h.

Corollary 5.2 (Uniform-center SK anti-concentration)**.** *For every finite n, in the zero-field Gaussian model,*

$$Q_{F_n}(h) \le \min\left\{1, \frac{2e^2(h+1)}{\sqrt{\mathrm{Var}\, F_n + 1}}\right\}, \qquad h \ge 0. \tag{12}$$

Conditional on the source variance lower bound, for every fixed $\eta > 0$, all sufficiently large n satisfy

$$Q_{F_n}(h) \le \min\{1, 2e^2(h+1)n^{-1/6+\eta}\} \quad \textit{simultaneously for all } h \ge 0.$$

The interval center may depend arbitrarily on n.

Proof. For completeness we recall the elementary bound used in the source: a log-concave probability density of standard deviation σ has supremum $M \le 2e^2/\sigma$. Choose a mode x_0. On either side, the density must reach M/e within distance e/M, or reach a support endpoint. Otherwise its integral on that side exceeds one. The decreasing secant slope of the log density then gives

$$q(x_0 + y) \le eMe^{-M|y|/e}.$$

Thus $\sigma^2 \le \mathbb{E}(Z - x_0)^2 \le 4e^4/M^2$. One-sided limits handle a mode at a support endpoint. Apply this to Proposition 4.1 and then Theorem 5.1 with rate one. □

This is stronger in scope than a statement only about a window centered at $\mathbb{E}F_n$: it controls every moving window, and fixed widths require no added logarithmic enlargement. The new element is the exact inverse-convolution interface, not the classical density bound, and it remains conditional on an actual variance certificate when used quantitatively for large n. It gives neither a limiting density nor a bound on derivatives of the original free-energy law.

6 A finite positive-Laplace certificate

The asymptotic window in the source has unspecified constants and threshold. It is not by itself a numerical certificate for a selected finite n. The following version makes that distinction precise.

Lemma 6.1 (Explicit one-dimensional constants)**.** *Let X be centered with a log-concave density and variance $\sigma^2 > 0$. Put*

$$h_0 = \frac{\log 3}{4}, \qquad b_0 = \frac{h_0^2}{16\sqrt{3}}, \qquad p_0 = \frac{h_0^4}{768}.$$

For $t > 0$, with $r = t\sigma/h_0$, one has

$$K_X(t) \le \frac{2r^2}{1-r} \quad (0 < r < 1), \tag{13}$$

$$K_X(t) \ge b_0 t\sigma + \log p_0. \tag{14}$$

The second inequality also holds if the moment is infinite.

Proof. The survival functions of X and $-X$ are log-concave. Chebyshev gives survival at -2σ at least $3/4$, and at 2σ at most $1/4$. The logarithmic secant gives $\mathbb{P}(X \ge u\sigma) \le (\sqrt{3}/4)e^{-h_0 u}$ for $u \ge 2$. Apply this to both signs and use the trivial bound for $u \le 2$; then $\mathbb{P}(|X| \ge u\sigma) \le 2e^{-h_0 u}$ for all $u \ge 0$. Integration yields $\mathbb{E}|X|^k \le 2k!(\sigma/h_0)^k$. Absolute convergence of the exponential series for $r < 1$, with the linear term zero, proves (13).

In particular $\mathbb{E}X^4 \le 48\sigma^4/h_0^4$. Interpolation between the first, second, and fourth absolute moments gives $\mathbb{E}|X| \ge h_0^2\sigma/(4\sqrt{3})$. Hence $\mathbb{E}X_+ \ge a_0\sigma$, where $a_0 = h_0^2/(8\sqrt{3})$, while $\mathbb{E}X_+^2 \le \sigma^2$. Splitting the first moment at $a_0\sigma/2$ and using Cauchy–Schwarz shows $\mathbb{P}(X \ge a_0\sigma/2) \ge a_0^2/4 = p_0$. Restriction of the exponential integral to this event proves (14). □

Theorem 6.2 (Variance and interval enclosures from one window)**.** *Suppose for an actual completed SK model, at a specified $0 < t < 1$, a rigorous enclosure is available:*

$$\widehat{K} - \varepsilon \le K_{F_n^c}(t) \le \widehat{K} + \varepsilon, \qquad \varepsilon \ge 0.$$

Set

$$L = \widehat{K} - \varepsilon - Tt^2/4 - t - \log(1-t), \qquad U = \widehat{K} + \varepsilon - Tt^2/4 - t - \log(1-t).$$

If $L > 0$, define

$$R(L) = \frac{\sqrt{L^2 + 8L} - L}{4} = \frac{2L}{\sqrt{L^2 + 8L} + L}, \quad \sigma_- = \frac{h_0 R(L)}{t}, \quad \sigma_+ = \frac{U - \log p_0}{b_0 t}.$$

Then

$$\max\{0, \sigma_-^2 - 1\} \le \operatorname{Var} F_n \le \sigma_+^2 - 1, \tag{15}$$

$$Q_{F_n}(h) \le \min\{1, 2e^2(h+1)/\max(1, \sigma_-)\}. \tag{16}$$

Several certified windows may be intersected by taking the largest lower and smallest upper standard-deviation bound.

Proof. Apply Lemma 6.1 to $Z_n - \mathbb{E}Z_n$ and use (9). If $r < 1$, $L \le 2r^2/(1-r)$ implies $r \ge R(L)$ by solving the quadratic inequality. If $r \ge 1$, the same conclusion follows from $0 < R(L) < 1$. The lower log-moment inequality and $K_{Z_n}(t) \le U$ give $\sigma_n \le \sigma_+$. Since $\sigma_n^2 = \operatorname{Var} F_n + 1$, both variance and concentration bounds follow. Valid premises automatically force $\sigma_+ \ge 1$. □

As an algebraic fixture, $L = 1$ gives $R(L) = 1/2$. At $t = 1/100$, the lower standard-deviation certificate is $50h_0$; the original variance bound is $2500h_0^2 - 1$, not the same quantity minus $1 + T/2$. This fixture does not assert that any finite SK instance has that certified window. The constants in the upper enclosure are deliberately conservative, whereas the unsmoothing coefficient itself is sharp.

For the source window $t = n^{-1/6+5m}$, $K_{F_n^c}(t) \in [c_{T,m}n^{20m}, C_{T,m}n^{100m}]$, the exact corrections in (9) are $O(t^2)$. Since $R(L) \to 1$ as $L \to \infty$, the resulting standard deviation lies between constant multiples of $n^{1/6-5m}$ and $n^{1/6+95m}$. Taking fixed $m > 0$ sufficiently small for a desired exponent slack reproduces, but does not improve, the source exponent. One may not take $m = m(n)$ in a theorem whose constants depend on fixed m.

7 Approximation errors must follow the actual law

The preceding certificates require rigorous enclosures, not substituted empirical moments. Two elementary stability statements clarify what kind of enclosure is sufficient.

Proposition 7.1 (Centered log-moment error)**.** *Let $F, \widetilde{F}$ be defined on the same probability space, with finite means and positive exponential moments at $t > 0$. Suppose $a \le \Delta := F - \widetilde{F} \le b$ almost surely, where a, b are finite constants, and write $d = \mathbb{E}\Delta$. Then*

$$t(a-d) \le K_F(t) - K_{\widetilde{F}}(t) \le t(b-d), \qquad |K_F(t) - K_{\widetilde{F}}(t)| \le t(b-a). \tag{17}$$

If $d \in [d_-, d_+]$ is certified, use $t(a-d_+)$ and $t(b-d_-)$ as the respective endpoints. For any finite sequence of normalized log-expectation operators $\mathcal{T}_m$, $m \ge 0$, uniform error intervals propagate unchanged through each exact operator. Additional stage errors add their interval endpoints. Finally, for square-integrable $F, \widetilde{F}$,

$$\left|\sqrt{\operatorname{Var} F} - \sqrt{\operatorname{Var} \widetilde{F}}\right| \le \|\Delta - \mathbb{E}\Delta\|_2 \le \|\Delta\|_2 . \tag{18}$$

Proof. With $\mathbb{E}_{\widetilde{F},t}$ denoting the probability measure proportional to $e^{t\widetilde{F}}$ times the original measure,

$$K_F(t) - K_{\widetilde{F}}(t) = \log \mathbb{E}_{\widetilde{F},t} e^{t\Delta} - t\mathbb{E}\Delta.$$

Bounding the integrand between e^{ta} and e^{tb} proves (17). Every $\mathcal{T}_m$ is order preserving and satisfies $\mathcal{T}_m(U+c) = \mathcal{T}_m U + c$, also at $m = 0$. Induction proves interval propagation without division by a small mass. The last assertion is the reverse triangle inequality for centered random variables in L^2, followed by the contraction of orthogonal projection onto the mean-zero subspace. □

A common deterministic offset cancels exactly in (17); it should not be counted twice as a centering error. On the other hand (18) gives no log-moment bound. For $X_M = M$ with probability $e^{-M/2}$ and zero otherwise, $\mathbb{E}X_M^2 \to 0$ but $\mathbb{E}e^{X_M} \to \infty$. Centering does not repair this example because $\mathbb{E}X_M \to 0$. Thus a small physical L^2 residual or agreement of finitely many polynomial moments cannot replace a tail-controlled exponential estimate.

The source comparison uses a per-spin fractional pressure p with exact normalization

$$K_{F_n^c}(s) = ns[p(T) - p(0)]. \tag{19}$$

If the two pressures have independent absolute error budgets ϵ_T, ϵ_0, the log-moment budget is $ns(\epsilon_T + \epsilon_0)$, not their unscaled sum. At $s = n^{-1/6+5m}$, making this negligible compared with the source lower window n^{20m} requires

$$\epsilon_T + \epsilon_0 = o(n^{-5/6+15m}).$$

Correlated endpoint errors can cancel, but such cancellation must itself be proved. This requirement is not satisfied merely by showing convergence of a discretization at each fixed n.

7.1 Fresh original Ram inputs

Three original local passages, rather than a later comparison catalogue, underlie the error discipline used here.

In the local-moment chapter [6], Taylor's formula is paired with integrals of actual monomials and an integral remainder. For a scalar function on a bounded interval this has the concrete form

$$\mathbb{E}f(X) = \sum_{j=0}^{k} \frac{f^{(j)}(c)}{j!}\mathbb{E}(X-c)^j + \mathcal{R}, \quad |\mathcal{R}| \le \frac{\sup|f^{(k+1)}|}{(k+1)!}\mathbb{E}|X-c|^{k+1}.$$

For exponential integrands the complement of that interval requires an absolute exponential-tail bound. The chapter's noncompact-domain and realizability distinctions prevent promotion of a polynomial identity to a closed finite description of $\mathcal{T}_m$. Our contribution is not a Taylor algorithm for the companion law.

The common-law covariance and remainder arguments of [7] require all quantities to use the same joint distribution. They explain the utility of the physical covariance decomposition in (3): positivity is retained before any estimate, and an approximation residual is not an independently addable noise variance. The square-root variance bound (18) is the corresponding scalar specialization.

The quotient differentiation proof in [8] retains the derivative of the denominator and the second quotient terms. For the positive normalized expectation in this report this becomes the covariance term in (2). Treating a tilted expectation as if its measure were fixed would erase that term and change the susceptibility. These are classical functional and moment principles with actual original proof support, not new claims about the spin-glass hierarchy.

8 Source architecture and inherited obligations

All thirteen cached texts have been literally read, including every scientific section, the bibliography, and the proof-order figure code. Reading a proof is not the same as independently certifying every estimate in it. The source's mechanism and the limits of this report are as follows.

Foundations (861 lines). Finite positive hierarchies are defined by normalized real-mass log expectations, not by analytic continuation in a replica number. The pinned posterior Hessian is a positive block matrix; its inverse controls the single-path energy by Brascamp–Lieb. Gauge transitivity then returns to the full unconditioned path law. Signed replica allocations retain fork atoms, terminal coincidences, and the complete topology coefficient. Cancellation is made before absolute values and only when the remaining factor is independent of the row placement. Fixed-size monotone-clock limits retain both sides of each cut and the prescribed joint filling of matrix and field jumps. No uniform discretization rate in n is supplied.

Comparisons (1232 lines). Constrained quantiles have both inverse-density floor and cap. The changed clocks are $b_0 - e(r)$ and $h_0 + re(r)$, and the optimized envelope pays $\frac{1}{4}\int e(r)r^2$. Contact conditions, cap averages, and first variations depend on actual admissible paths. The bootstrap uses a deterministic good split chosen before moment order or cavity depth. The present Theorem 3.3 modifies its one-branch projection step only; Section 3.1 tracks that change in the off-diagonal pair error. Whole-cap traces do not become pointwise cap bounds, and small floor mass does not imply small floor radius.

Cavity (1222 lines). Deleting one site retains the denominator n and all undeleted noise covariances. The scalar comparison uses the entire path $K = h + bS$, not just one endpoint value. Newly generated signed row coefficients and scalar centers stay fixed during that row's interpolation. Observed blocks and unobserved attachments have different parity and contraction roles. Susceptibility is calibrated at a deterministic regular band; the finite longitudinal recursion is fixed before $n \to \infty$. This report neither re-proves that recursion nor allows its depth or degree to grow with n.

Profile (913 lines). The scalar PDE is

$$V_q = -\frac{T}{2}(V_{zz} + \alpha V_z^2), \qquad V(1, z) = \log\cosh z,$$

with its specified diffusion. The identities for $\Gamma' = T\mathbb{E}V_{zz}^2$ and $\Gamma'' = T^2\mathbb{E}(V_{zzz}^2 - 2\alpha V_{zz}^3)$ lead to cubic small-clock closure and a gap-exclusion argument in the virtual potential. Both mass and radius errors are used. The macroscopic limit is a variational comparison to the Parisi minimizer, with an explicit square-completion penalty $e(T-e)(r-z)^2/T$. Full support near zero is imported from Lopatto's theorem; it is not deduced from finite moments in this report.

Propagation (691 lines). A failed profile at a selected outermost scale creates a stronger child comparison. A successful child error would force profile agreement and contradict that failure; otherwise the next error parameter grows by a fixed power of n. Matrix margins are reserved before a finite chain is chosen. The argument distinguishes an admissible bare endpoint from a fixed-strength root whose bare coefficients may not define a covariance path. It does not evaluate the latter pressure. Our fixed-n subdivision inside one branch is unrelated to this fixed-depth contradiction.

Fluctuations (702 lines). The comparison is applied at $s = n^{-1/6+5m}$ to obtain the positive window with exponents $20m$ and $100m$. A high-only field wall controls the omitted low group; its coefficient, matrix boundary gap, and physical scale are retained during the sweep. Exponential augmentation and log-concavity then convert the positive window to variance and typical scale. We preserve those mechanisms. Sections 5–6 refine the distributional interface, not the wall construction.

Companion interface (292 lines). Mass rescaling to a bounded terminal mass changes Hessian bounds and attachment weights in a specified way. Physical size remains in every overlap denominator. Conditional replacement requires the complete specified prefix and exterior, with no unrevealed target future. Clock-independent identities alone do not permit an unbounded row-covariance transfer. A sequential common temperature limit $T_n \to T_\infty > 1$ keeps T_n fixed throughout each size's chain. The appendix explicitly does not assert uniformity at the critical temperature or for growing recursion depth.

Introduction and ancillary texts. The introduction (206 lines) states the exponent theorem and proof dependencies; the 58-line paper driver and one-line main driver specify the module order. The 28-line figure records the two child-success/failure branches. The 333-line bibliography and 18-line README supply references and provenance. These six ancillary/introduction texts together with the seven scientific modules make thirteen files.

9 Current primary literature and status

The public refresh was made on October 9, 2026. Aronow–Lopatto's September 8 preprint [2] states fixed low-temperature variance bounds with powers 4/15 and 7/15. Its transitive Gaussian Proposition 5.1 already covers augmentation of the off-diagonal free energy. The source under study claims the sharper exponent 1/3 for variance; publication of a cached manuscript is not independent verification of that claim.

The source cites Lopatto's version 3 for full interval support. The current primary record is version 4, dated October 5 [3]. Its Theorem 1.1 states interval support, a smooth density extending to the endpoints, and an atom at the upper endpoint for every fixed $\beta > 1$. The statement was checked; its full external proof was not audited here. Thus source uses of this theorem remain explicitly imported.

Du–Huang's critical-temperature preprint [4], version 2 of July 8, states variance $\frac{1}{6}\log n + O(1)$ and a Gaussian central limit theorem at $\beta = 1$. That is a different regime and supplies no low-temperature uniformity. Prékopa's classical marginal theorem [5] is the structural analytic input in the augmentation argument. None of these facts supplies a limiting law at fixed $\beta > 1$ from the present local estimates.

10 Advantages, costs, and research directions

Interface	Demonstrated benefit	Cost or unresolved gate
Planted projection	Removes $(\log n)^2$ by exact allocation; off-diagonal budget loses four fewer logarithms	Every finite subdivision must preserve the actual planted future; constants still depend on fixed moment order
Exponential inverse	Sharp $h+1$ interval coefficient at rate one; all centers, including atoms and fixed widths	Needs a density supremum bound; variance input is not produced by finite moment fixtures
Positive-Laplace window	Explicit standard-deviation and anti-concentration enclosures	Actual finite-n log-moment bounds, not unspecified eventual constants, are required
Normalized error transfer	Common offsets cancel and stage intervals add without small-mass division	Uniform/tail-certified errors can be expensive; an L^2 bound does not control an exponential tilt

The projection refinement removes an artifact of equal subdivisions; it does not change the physical field-clock budget or the final power exponent. An implementable discretization has finitely many resolvable windows and should use (6), with its actual smallest clock length and remainder, rather than the limiting formula alone. Allowing nonconstant certified one-step coefficients would lead to the same weighted optimization with those coefficients retained. Establishing uniform coefficients under altered conditioned laws is a separate problem.

For distributional work, the useful next step is a rigorously quantified finite positive window at named n, β, t, including the comparison and field-wall errors. Optimizing the coarse one-dimensional constants can improve numerical utility without changing any exponent. Extending augmentation to a model without transitive gauge symmetry needs a new structural proof. Varying temperature, non-Gaussian disorder, a limiting law, and a variance prefactor remain distinct research questions.

11 Evidence, corrections, and assessment

The public read covers 6557 literal lines across thirteen cached texts. Fresh original-private reads cover 846 distinct lines: the full 446-line local-moment chapter; lines 1–52 and 216–328 of the 939-line common-law covariance report; and lines 1–98, 328–426, and 2104–2141 of the 2250-line normalized-functional report. Child inputs and all complementary private lines are unread. No whole private corpus or full external bibliography is claimed read. The owned ledger stores pre-read hashes, actual display intervals, post-read hashes, and unread complements. Snapshot provenance does not assert the state of a live branch.

The independent author program uses exact rational arithmetic. It passes 3174 parameter cases and 12978 assertions: 96 finite allocations, 6 limiting-constant fixtures, 2475 exponential interval inversions, 24 sharp plateau fixtures, 435 Laplace-root cases, 90 positive spin-law covariance cases, and 48 normalization/error cases. The latter spin laws are deliberately identified as algebraic fixtures, not simulations of the required planted SK hierarchy. Atoms at both interval endpoints are included. These tests support the local algebra and examples, not the global exponent theorem; independent external review counts are not included.

No source theorem is declared false by this report. Valid source details retained include the completed covariance, real masses, fork atoms, physical overlap denominator, exterior-anchor gate, fixed-depth order of limits, and the two distinct cap/floor errors. The refinements are the optimized clock allocation and the sharp one-sided inverse interface. The original-model augmentation and

elementary log-concave estimates are credited as existing tools; the completed-variable and original-variable variances are not interchanged.

The C3 assessment is based on a proved removal of a source logarithmic loss with downstream error accounting, together with a sharp quantitative anti-concentration interface at all centers. It is independent of the catalogue's applicability level L3. It does not claim historical novelty for weighted optimization, Stieltjes inversion, or log-concavity, and it is not credit for a new global spin-glass theorem.

12 Conclusion

The most useful improvements here occur at interfaces: allocating a fixed field interval according to the actual contraction weights, and undoing a one-sided exponential convolution without a probabilistic truncation loss. Both have exact finite formulas and transparent limitations. They strengthen the source's local quantitative machinery while leaving its global overlap, profile, and propagation obligations visible.

References

[1] OpenAI, *The low-temperature Sherrington–Kirkpatrick fluctuation scale*, September 24, 2026. Pinned cached build; public source release directory.

[2] P. M. Aronow and P. Lopatto, *Quantitative Parisi formulas and fluctuations in the Sherrington–Kirkpatrick model*, September 8, 2026, arXiv:2609.09103v1.

[3] P. Lopatto, *Full replica symmetry breaking in the Sherrington–Kirkpatrick model*, version 4, October 5, 2026, arXiv:2607.11756v4.

[4] H. Du and B. Huang, *Fluctuations of the Sherrington–Kirkpatrick free energy at critical temperature*, version 2, July 8, 2026, arXiv:2607.02172v2.

[5] A. Prékopa, *On logarithmic concave measures and functions*, Acta Sci. Math. (Szeged) **34** (1973), 335–343. Primary repository record.

[6] M. Subbarao, *Ram Calculus*, volume 1, revision 2, chapter 3, local moments. Confidential original, local file `ch03-local-moments-rev2.tex`; literal scope stated above.

[7] M. Subbarao, *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16, October 1, 2026. Confidential original, scoped common-law covariance and remainder proofs.

[8] M. Subbarao, *Urysohn-Ram Equations and State-Dependent Kernels: Functional and Spatial Derivatives, Moving Supports, Nonlinear Panels, Calibration, and Complex Systems*, Report 6, September 18, 2026. Confidential original, scoped normalized-quotient proofs.